```
import { useState } from "react";

const PLATFORMS = [
  { id: "macos", label: "macOS", icon: "⌘", color: "#a8d8ea" },
  { id: "ios", label: "iOS", icon: "📱", color: "#b8e0d2" },
  { id: "android", label: "Android", icon: "🤖", color: "#d6eadf" },
];

const DEVICES = {
  macos: [
    { tier: "FLOOR", name: "MacBook Air M1", chip: "M1", ram: "8GB", os: "macOS 1
    { tier: "MID", name: "MacBook Pro M2", chip: "M2", ram: "16GB", os: "macOS 13
    { tier: "HIGH", name: "MacBook Pro M3 Pro", chip: "M3 Pro", ram: "36GB", os:
  ],
  ios: [
    { tier: "FLOOR", name: "iPhone 12", chip: "A14 Bionic", ram: "4GB", os: "iOS
    { tier: "MID", name: "iPhone 14", chip: "A15 Bionic", ram: "6GB", os: "iOS 16
    { tier: "HIGH", name: "iPhone 15 Pro", chip: "A17 Pro", ram: "8GB", os: "iOS
  ],
  android: [
    { tier: "FLOOR", name: "Pixel 6a", chip: "Tensor G1", ram: "6GB", os: "Androi
    { tier: "MID", name: "Galaxy A54", chip: "Exynos 1380", ram: "8GB", os: "Andr
    { tier: "HIGH", name: "Galaxy S23", chip: "Snapdragon 8 Gen 2", ram: "8GB", o
  ],
};

const CAPABILITIES = [
  {
    category: "🧠 LLM Inference",
    items: [
      {
        id: "llm_cpu",
        name: "LLM — CPU only (Q4 1B)",
        desc: "Llama 3.2 1B via llama.cpp, no GPU",
        expected: ">8 tok/s",
        macos: { floor: "✅", mid: "✅", high: "✅" },
        ios: { floor: "✅", mid: "✅", high: "✅" },
        android: { floor: "✅", mid: "✅", high: "✅" },
        priority: "P0",
      },
      {
        id: "llm_gpu",
        name: "LLM — GPU accelerated (Q4 1B)",
        desc: "Metal on Apple / Vulkan on Android",
```

```
      expected: ">15 tok/s",
      macos: { floor: "✅ Metal", mid: "✅ Metal", high: "✅ Metal" },
      ios: { floor: "✅ Metal", mid: "✅ Metal", high: "✅ Metal" },
      android: { floor: "✅ Vulkan", mid: "✅ Vulkan", high: "✅ Vulkan" },
      priority: "P0",
    },
    {
      id: "llm_3b",
      name: "LLM — Phi-3.5 Mini (Q4 3B)",
      desc: "Larger model, needs ≥6GB RAM",
      expected: ">10 tok/s",
      macos: { floor: "⚠️ 8GB tight", mid: "✅", high: "✅" },
      ios: { floor: "❌ OOM risk", mid: "✅", high: "✅" },
      android: { floor: "⚠️ borderline", mid: "✅", high: "✅" },
      priority: "P1",
    },
    {
      id: "llm_stream",
      name: "Streaming token output",
      desc: "Dart stream API, token-by-token UI",
      expected: "No jank",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
    },
    {
      id: "llm_json",
      name: "JSON mode output",
      desc: "Constrained grammar output",
      expected: "Valid JSON",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P1",
    },
  ],
},
{
  category: "🎤 Speech-to-Text (STT)",
  items: [
    {
      id: "stt_tiny",
      name: "Whisper Tiny — transcription",
      desc: "Short audio <10s, whisper.cpp",
      expected: "<500ms",
      macos: { floor: "✅", mid: "✅", high: "✅" },
```

```
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
    },
    {
      id: "stt_realtime",
      name: "Real-time streaming STT",
      desc: "Mic input → live transcription via VAD",
      expected: "<200ms lag",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "⚠️ VAD tuning", mid: "✅", high: "✅" },
      priority: "P1",
    },
    {
      id: "stt_multilang",
      name: "Multi-language detection",
      desc: "Auto-detect non-English input",
      expected: "Correct lang",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P2",
    },
  ],
},
{
  category: "🔊 Text-to-Speech (TTS)",
  items: [
    {
      id: "tts_basic",
      name: "Kokoro-82M basic synthesis",
      desc: "Text → audio playback",
      expected: "<100ms TTFA",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
    },
    {
      id: "tts_pitch",
      name: "Pitch / rate adjustment",
      desc: "Real-time parameter changes",
      expected: "Smooth output",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "⚠️ test NNAPI", mid: "✅", high: "✅" },
```

```
      priority: "P2",
    },
  ],
},
{
  category: "⚙️ Hardware Acceleration",
  items: [
    {
      id: "hw_metal",
      name: "Metal GPU path",
      desc: "Apple GPU inference, not CPU fallback",
      expected: "GPU active",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "N/A", mid: "N/A", high: "N/A" },
      priority: "P0",
    },
    {
      id: "hw_vulkan",
      name: "Vulkan GPU path",
      desc: "Android GPU inference",
      expected: "GPU active",
      macos: { floor: "N/A", mid: "N/A", high: "N/A" },
      ios: { floor: "N/A", mid: "N/A", high: "N/A" },
      android: { floor: "✅ Mali", mid: "✅ Exynos", high: "✅ Adreno" },
      priority: "P0",
    },
    {
      id: "hw_nnapi",
      name: "NNAPI delegation",
      desc: "Android Neural Networks API",
      expected: "NPU utilised",
      macos: { floor: "N/A", mid: "N/A", high: "N/A" },
      ios: { floor: "N/A", mid: "N/A", high: "N/A" },
      android: { floor: "⚠️ limited", mid: "⚠️ partial", high: "✅ full" },
      priority: "P1",
    },
    {
      id: "hw_cpu_fallback",
      name: "CPU fallback (GPU unavailable)",
      desc: "Graceful downgrade when GPU fails",
      expected: "No crash",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
    },
```

```
  ],
},
{
  category: "📦 Model Management",
  items: [
    {
      id: "model_load",
      name: "Model load from local storage",
      desc: "GGUF file parse + mmap",
      expected: "<3s cold load",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
    },
    {
      id: "model_download",
      name: "OTA model download + checksum",
      desc: "Progressive download + resume",
      expected: "SHA verified",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P1",
    },
    {
      id: "model_memory",
      name: "Memory watchdog under pressure",
      desc: "OS kills app if over RAM limit",
      expected: "Graceful error",
      macos: { floor: "⚠️ 8GB limit", mid: "✅", high: "✅" },
      ios: { floor: "⚠️ 4GB tight", mid: "✅", high: "✅" },
      android: { floor: "⚠️ monitor", mid: "✅", high: "✅" },
      priority: "P0",
    },
  ],
},
{
  category: "🔋 Performance & Battery",
  items: [
    {
      id: "perf_thermal",
      name: "Thermal throttling sustained load",
      desc: "10-min continuous inference",
      expected: "No crash/throttle",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "⚠️ 12 gets hot", mid: "✅", high: "✅" },
```

```
      android: { floor: "⚠️ Tensor warm", mid: "⚠️ Exynos warm", high: "✅" },
      priority: "P1",
    },
    {
      id: "perf_battery",
      name: "Battery drain per 100 tokens",
      desc: "mAh measurement over inference",
      expected: "Baseline logged",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P2",
    },
    {
      id: "perf_background",
      name: "Background / foreground state",
      desc: "App backgrounded mid-inference",
      expected: "Pause + resume",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "⚠️ iOS kills BG", mid: "⚠️", high: "⚠️" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P1",
    },
  ],
},
{
  category: "🌐 Offline & Privacy",
  items: [
    {
      id: "offline_airplane",
      name: "Airplane mode inference",
      desc: "Full inference, zero network",
      expected: "100% functional",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
    },
    {
      id: "offline_network_none",
      name: "No data leaves device",
      desc: "Charles proxy — verify 0 outbound",
      expected: "0 requests",
      macos: { floor: "✅", mid: "✅", high: "✅" },
      ios: { floor: "✅", mid: "✅", high: "✅" },
      android: { floor: "✅", mid: "✅", high: "✅" },
      priority: "P0",
```

```
      },
    ],
  },
];

const TIER_COLORS = {
  FLOOR: "#f0a500",
  MID: "#3a86ff",
  HIGH: "#06d6a0",
};

const PRIORITY_COLORS = {
  P0: { bg: "#ff4757", text: "#fff" },
  P1: { bg: "#ffa502", text: "#fff" },
  P2: { bg: "#5352ed", text: "#fff" },
};

function statusStyle(val) {
  if (!val || val === "N/A") return { color: "#555", opacity: 0.5 };
  if (val.startsWith("✅")) return { color: "#06d6a0", fontWeight: 600 };
  if (val.startsWith("⚠️")) return { color: "#ffa502", fontWeight: 600 };
  if (val.startsWith("❌")) return { color: "#ff4757", fontWeight: 600 };
  return { color: "#ccc" };
}

export default function EdgeVedaMatrix() {
  const [activePlatform, setActivePlatform] = useState("all");
  const [activeCategory, setActiveCategory] = useState("all");
  const [activePriority, setActivePriority] = useState("all");

  const filteredCaps = CAPABILITIES
    .map(cat => ({
      ...cat,
      items: cat.items.filter(item => {
        if (activePriority !== "all" && item.priority !== activePriority) return
        return true;
      }),
    }))
    .filter(cat =>
      (activeCategory === "all" || cat.category === activeCategory) && cat.items.
    );

  const showPlatform = (pid) => activePlatform === "all" || activePlatform === pi

  return (
    <div style={{
      fontFamily: "'JetBrains Mono', 'Fira Code', monospace",
```

```
  background: "#0a0e1a",
  minHeight: "100vh",
  color: "#e2e8f0",
  padding: "32px 24px",
}}>
  {/* Header */}
  <div style={{ marginBottom: 36 }}>
    <div style={{ display: "flex", alignItems: "center", gap: 12, marginBotto
      <div style={{
        background: "linear-gradient(135deg, #a8d8ea, #06d6a0)",
        borderRadius: 8,
        padding: "6px 12px",
        fontSize: 11,
        fontWeight: 700,
        color: "#0a0e1a",
        letterSpacing: 2,
        textTransform: "uppercase",
      }}>Edge Veda SDK</div>
      <div style={{ color: "#4a5568", fontSize: 11 }}>v1.0 · Physical Device
    </div>
    <h1 style={{
      fontSize: 28,
      fontWeight: 800,
      margin: 0,
      background: "linear-gradient(90deg, #a8d8ea 0%, #06d6a0 50%, #b8e0d2 10
      WebkitBackgroundClip: "text",
      WebkitTextFillColor: "transparent",
      letterSpacing: -1,
    }}>Capability × Platform × Device Matrix</h1>
    <p style={{ color: "#718096", fontSize: 12, marginTop: 8 }}>
      CPU/GPU AI Inference · Flutter · macOS · iOS · Android · Physical Devic
    </p>
  </div>

  {/* Device Cards */}
  <div style={{
    display: "grid",
    gridTemplateColumns: "repeat(3, 1fr)",
    gap: 16,
    marginBottom: 32,
  }}>
    {PLATFORMS.map(p => (
      <div key={p.id} style={{
        background: "#111827",
        border: "1px solid #1f2937",
        borderRadius: 12,
        padding: 16,
```

```
        borderTop: `3px solid ${p.color}`,
      }}>
        <div style={{ fontSize: 13, fontWeight: 700, color: p.color, marginBo
          {p.icon} {p.label}
        </div>
        {DEVICES[p.id].map(d => (
          <div key={d.name} style={{
            marginBottom: 10,
            paddingBottom: 10,
            borderBottom: "1px solid #1f2937",
          }}>
            <div style={{ display: "flex", alignItems: "center", gap: 6, marg
              <span style={{
                fontSize: 9,
                fontWeight: 700,
                padding: "2px 6px",
                borderRadius: 4,
                background: TIER_COLORS[d.tier] + "22",
                color: TIER_COLORS[d.tier],
                letterSpacing: 1,
              }}>{d.tier}</span>
              <span style={{ fontSize: 12, fontWeight: 700, color: "#e2e8f0"
            </div>
            <div style={{ display: "flex", gap: 12, fontSize: 10, color: "#71
              <span>🔲 {d.chip}</span>
              <span>💾 {d.ram}</span>
              <span>🖥️ {d.os}</span>
            </div>
            <div style={{ fontSize: 10, color: "#4a5568", marginTop: 3 }}>↳ {
          </div>
        ))}
      </div>
    ))}
  </div>

  {/* Filters */}
  <div style={{ display: "flex", gap: 12, flexWrap: "wrap", marginBottom: 24,
    <span style={{ fontSize: 11, color: "#4a5568", letterSpacing: 1, textTran

    {/* Platform filter */}
    <div style={{ display: "flex", gap: 6 }}>
      {["all", ...PLATFORMS.map(p => p.id)].map(pid => (
        <button key={pid} onClick={() => setActivePlatform(pid)} style={{
          padding: "4px 10px", borderRadius: 6, fontSize: 11, fontWeight: 600
          background: activePlatform === pid ? "#3a86ff" : "#1f2937",
          color: activePlatform === pid ? "#fff" : "#718096",
          border: "none", letterSpacing: 0.5,
```

```
        }}>{pid === "all" ? "All Platforms" : pid.toUpperCase()}</button>
      ))}
    </div>

    <span style={{ color: "#2d3748" }}>|</span>

    {/* Priority filter */}
    {["all", "P0", "P1", "P2"].map(p => (
      <button key={p} onClick={() => setActivePriority(p)} style={{
        padding: "4px 10px", borderRadius: 6, fontSize: 11, fontWeight: 600,
        background: activePriority === p
          ? (p === "all" ? "#4a5568" : PRIORITY_COLORS[p].bg)
          : "#1f2937",
        color: activePriority === p ? "#fff" : "#718096",
        border: "none",
      }}>{p === "all" ? "All Priorities" : p}</button>
    ))}
  </div>

  {/* Legend */}
  <div style={{ display: "flex", gap: 20, marginBottom: 20, fontSize: 11, col
    <span>✅ Supported</span>
    <span>⚠️ Partial / Risk</span>
    <span>❌ Not supported</span>
    <span style={{ color: "#4a5568" }}>N/A Not applicable</span>
    <span style={{ marginLeft: "auto", display: "flex", gap: 8 }}>
      {["P0", "P1", "P2"].map(p => (
        <span key={p} style={{
          padding: "2px 6px", borderRadius: 4, fontSize: 10, fontWeight: 700,
          background: PRIORITY_COLORS[p].bg, color: PRIORITY_COLORS[p].text,
        }}>{p} {p === "P0" ? "Critical" : p === "P1" ? "Important" : "Nice-to
      ))}
    </span>
  </div>

  {/* Matrix Table */}
  {filteredCaps.map(cat => (
    <div key={cat.category} style={{ marginBottom: 32 }}>
      <div style={{
        fontSize: 13, fontWeight: 700, color: "#a8d8ea", marginBottom: 10,
        letterSpacing: 1, textTransform: "uppercase", borderLeft: "3px solid
        paddingLeft: 10,
      }}>{cat.category}</div>

      <div style={{ overflowX: "auto" }}>
        <table style={{ width: "100%", borderCollapse: "collapse", fontSize:
          <thead>
```

```
  <tr style={{ background: "#111827" }}>
    <th style={{ textAlign: "left", padding: "8px 12px", color: "#4
    <th style={{ textAlign: "left", padding: "8px 12px", color: "#4
    <th style={{ padding: "4px", color: "#4a5568", fontWeight: 600,
    {PLATFORMS.filter(p => showPlatform(p.id)).map(p => (
      ["FLOOR", "MID", "HIGH"].map(tier => (
        <th key={`${p.id}-${tier}`} style={{
          textAlign: "center", padding: "6px 8px",
          color: TIER_COLORS[tier], fontWeight: 700, fontSize: 10,
        }}>
          <div style={{ opacity: 0.7, fontSize: 9, marginBottom: 2
          {tier}
        </th>
      ))
    ))}
  </tr>
</thead>
<tbody>
  {cat.items.map((item, i) => (
    <tr key={item.id} style={{
      background: i % 2 === 0 ? "#0d1117" : "#111827",
      borderBottom: "1px solid #1a2235",
    }}>
      <td style={{ padding: "10px 12px" }}>
        <div style={{ fontWeight: 700, color: "#e2e8f0", marginBott
        <div style={{ color: "#4a5568", fontSize: 10 }}>{item.desc}
      </td>
      <td style={{ padding: "10px 12px", color: "#718096", fontSize
      <td style={{ padding: "8px", textAlign: "center" }}>
        <span style={{
          padding: "2px 5px", borderRadius: 3, fontSize: 9, fontWei
          background: PRIORITY_COLORS[item.priority].bg,
          color: PRIORITY_COLORS[item.priority].text,
        }}>{item.priority}</span>
      </td>
      {PLATFORMS.filter(p => showPlatform(p.id)).map(p => (
        ["floor", "mid", "high"].map(tier => {
          const val = item[p.id]?.[tier] || "—";
          return (
            <td key={`${p.id}-${tier}`} style={{
              textAlign: "center", padding: "8px",
              borderLeft: tier === "floor" ? "1px solid #1a2235" :
              ...statusStyle(val),
            }}>
              <span style={{ fontSize: 11 }}>{val}</span>
            </td>
          );
```

```
                    })
                  ))}
                </tr>
              ))}
            </tbody>
          </table>
        </div>
      </div>
    ))}

    {/* Summary Stats */}
    <div style={{
      marginTop: 24, padding: 16, background: "#111827", borderRadius: 12,
      border: "1px solid #1f2937", display: "flex", gap: 32, flexWrap: "wrap",
    }}>
      {[
        { label: "Total Capabilities", value: CAPABILITIES.reduce((a, c) => a +
        { label: "P0 Critical Tests", value: CAPABILITIES.reduce((a, c) => a +
        { label: "Platforms", value: 3 },
        { label: "Test Devices", value: 9 },
        { label: "Physical Device Only", value: "100%" },
      ].map(s => (
        <div key={s.label}>
          <div style={{ fontSize: 22, fontWeight: 800, color: "#06d6a0" }}>{s.v
          <div style={{ fontSize: 10, color: "#4a5568", letterSpacing: 0.5 }}>{
        </div>
      ))}
    </div>

    <div style={{ marginTop: 16, fontSize: 10, color: "#2d3748", textAlign: "ce
      Edge Veda SDK · edge-veda · pub.dev · github.com/ramanujammv1988/edge-ved
    </div>
  </div>
);
}
```